Lab 2: Basic Robotics 2 – DEIS DT8007

Supervisor: Martin Cooney (~ 2 hours)

Goal: after completing this lab, you will be able to:

- (a) Write some simple programs for a microcontroller
- (b) Do some basic hardware prototyping
- (c) Fly a small drone
- (d) Get a small ground robot to move

Motivation: Designers of embedded and intelligent systems often have to program microcontrollers (e.g., to be "robot brains"), modify and check hardware, and get various kinds of systems to run, like robots. Microcontrollers are immensely useful since they are cheap, small, require sometimes very little power, and can be programmed to do a vast range of tasks, from reading sensors to controlling actuators. Arduinos are very easy-to-use, cheap, versatile microcontrollers that are widely available and can help you to build robots and accomplish a huge variety of tasks. For example, the Arduino Uno R4 Wifi board has a ESP32 co-processor, LED matrix, CAN bus support, etc. It's also often needed to be able to modify and check hardware: e.g., via using a multimeter, drilling, soldering. Ryze Tello drones are very affordable, easy-to-use small drones that can carry a relatively substantial payload. Redbots are a nice out-of-the-box solution that easily allow a variety of sensors, from encoders, to infrared line sensors, accelerometers, and more.

All labs in this course relate mainly to the second intended learning outcome in the course syllabus, called "Skills and ability: program an embedded computer implementing intelligent functions for sensing (such as situation and position estimation, comprising calibration, filtering, and fusing signals from different sensors), actuator control, and wireless communication in a distributed system".

Note the labs in this course are separate from the project! They might sometimes also help you with the project but cover a wider range of different tasks that might not be part of the project.

Approach: to complete the lab, you will require the following:

- Adequate preparation. You should have downloaded this file for the lab from Blackboard and read it briefly before starting.
- A group.
- Access to:
  - The A113 project room
  - Your group's Arduino, Ryze Tello, and Redbot

Result:

- A submitted report. After the lab, please submit a short report in pdf format with filename "DEIS_labX_groupY.pdf" including screenshots and any comments, through Blackboard. (Replace "X" with the lab number and "Y" with your group number.) Please see our shared deadlines document to see the recommended dates of completion and absolute deadlines for this lab.

You will receive a response by email within 15 working days from the deadline at the latest. Your report will be judged as passing, reasonable but failing, or unsubmitted. Any feedback will typically be short, around a line or two in the case the lab is still failing, that should indicate what remains to be done to attain a passing grade.

Please note:

- If the report is reasonable but failing, you will get a chance to review and resubmit.
- All labs must be submitted to receive a passing grade in the course.
- It's okay to use AI tools somehow in this lab if you want, like ChatGPT or such; just you should declare it.
- Your report should include a table detailing exactly which group member did what. Please make sure to always be truthful, clear, and precise. It's perfectly fine to divide up labs so that not all group members work on them, but each group member should at least have seen the submitted report. A table can look like this (with names given and activities explained):

	Member1	Member2	Member3	Member4
Activity 1	50%	50%		
Activity 2		100%		
Activity 3	25%	25%	25%	25%

- **You are expected to do original work. Use of someone else's code/report/words/etc without referencing them is a _serious_ problem which will be reported to the disciplinary board.**

Tasks

My suggestion is that the group can allocate members to deal with each task set (a-d), then gather together afterwards to share information, discuss and compile the report.

(a) ARDUINO

1. Get started with the Arduino (~5 minutes)

Follow the instructions at the link below:

https://docs.arduino.cc/tutorials/uno-r4-wifi/r4-wifi-getting-started/

Download the latest version of the Arduino software onto some computer (e.g., a laptop).

Start it up.

Click the “boards manager” icon on the left-side menu (it’s the second gray icon from the top, that looks a bit like a microcontroller).

Search for “UNO R4 WiFi”, and click “install”

Connect your Arduino board to your computer via the cable.

Select the right board from the top menu:

Tools – Board – Arduino Uno R4 Boards - Arduino Uno R4 WiFi

Select the right port under Tools – Port

Great, now you should be able to use the Arduino with your computer!

Write a test sketch to your Arduino, like Blink.

Take a screenshot or photo of your Arduino running some sketch and add it to your report.

2. Connect to WiFi (~5 minutes)

Open File - Examples - WiFiS3 - SimpleWebServerWifi

Press arduino_secrets.h on the top tab.

Add the SSID and password for the course router.

Click the right arrow to upload the sketch to your Arduino board.

Open the Serial Monitor. Copy the IP address that is shown there.

Connect your laptop to our course’s router’s WiFi network if it’s not connected already.

Open an internet browser on your computer like Firefox.

Now paste the IP address above into the browser’s search field.

You will see two links to turn an LED on or off. Try it.

(You can also do this manually by accessing the addressed below:

http://PASTE_IPADDRESS_HERE/H

http://PASTE_IPADDRESS_HERE/L)

Great! Now you can control/access info from a microcontroller over WiFi!

Take a screenshot of the board with its LED on in front of the browser and add it your report.

(b) PROTOTYPING

Martin will go around the room, maybe in group order. Together we will do some simple tasks with multimeter, soldering, and drilling (about 10 min per group). Please consider watching the linked videos on Blackboard before the lab to refresh your memory. Get one group member to take a souvenir photo and include it in your report.

(c) DRONE

Ensure your group has their drone (and a charged drone battery). Each drone has a label with a ”name”.

Group 1: Gebo

Group 2: Fehu

Group 3: Uruz

Group 4: Wunjo

Group 5: Raido

1. Manual flying (~5 minutes)

Download the Ryze Tello App on your smartphone/device. You might need an older phone. If no one in your group can do download/run the app, Martin can lend his phone.

Slide a battery into the drone, turn on drone's power button, connect to its hotspot with your device (it will be TELLO something), and try controlling it manually via the app.

First you press/slide the button on the top left to get the drone to take off.

The first time you fly be extra careful: be ready to immediately/quickly land the drone. You will need to use the buttons on the top left _twice_ to get it to land. Be very careful to not collide with people/break anything. You will want a good amount of free space around the drone, and you can only pilot it in indoors!

If your drone has taken off and seems relatively stable:

Use the controls on the left first:

Get your drone to rotate left and right.

Get your drone to fly up and down.

If all worked well, use the control on the right to get your drone to fly forward and backward, and strafe left and right.

When done, land, turn off the drone by pressing its power button and verifying the lights vanish, then remove the battery.

If the drone drifts badly, it needs to be recallibrated using the Ryze Tello App. Do the full imu callibration, removing the arm guards and propellers with the special tool, then lay it on each side

following the instructions in the app. You get there by pressing the gear icon for settings, then on detailed settings.

If the drone is drifting just a little, you can also press the center of mass adjustment. Just let it hover for a minute or so. You will see a message when it's done.

Note: Attaching any payload right below the drone will mess with its IR sensor, leading to dangerous behavior like continuously banging into the ceiling, so generally it's better to add payloads to the top of the drone.

To charge the battery, there are two ways: Either (1) slide in a usb micro cable adapter into the side of the drone when the drone's power is off, or (2) use the charger with slots for 3 batteries in the project room.

(Don't leave batteries inside the drone when not in use. Always keep batteries charged; don't leave them lying around in a depleted state. Don't charge batteries when they are hot just after being used. Don't leave the batteries charging after they are done; when the green led lights, remove the batteries and charger.)

Get one group member to take a souvenir photo and include it in your report.

2. Control via python (~30 minutes)

Next after you have verified that everything is basically working (the drone can fly stably and you can see the camera feed): please try getting the video feed from the drone and controlling it from your own Python program on your computer. I suggest trying code from our wonderful volunteer a few years back, Shiva:

https://github.com/SHiTU23/Tello_drone/blob/main/src/control_and_capture.py

If that doesn't work, send me a message and I can send you my own (ancient/rough) code. There also used to be many other libraries online.

If this works, you can even go one step further and try conmmanding the drone autonomously. A good first autonomous program involves just rotating first rather than any kind of translations (e.g., take off, wait a sec, rotate 90 degrees, wait, rotate, wait, land). It's also best if your code allows any motions to be cancelled quickly.

Get one group member to take a souvenir photo and include it in your report. E.g., "s" with Shiva's code to save an image from the drone's camera.

(d) GROUND ROBOT

(~30 minutes)

Assemble your ground robot if it is not assembled already. This means mounting the red microcontroller board (it's basically an Arduino Uno with some extra stuff like motor drivers), IR sensors, encoders. For ease, you can just use the alkaline battery packs.

Complete the tutorials here: 1-3 basic motions, 6 line following, 7 encoders

https://learn.sparkfun.com/tutorials/experiment-guide-for-redbot-with-shadow-chassis)

Use some tape to make yourself a track somewhere for line following.

Get one group member to take a souvenir photo and include it in your report.